Lift

There are two common questions that come to mind.

- What causes lift force?
- How to predict the lift?

To predict the lift force very accurately we can use

- Thin–airfoil Theory
- Kutta – Joukowski Theorem
- Bernoulli's principle

Where does the lift come from?

Lift is a balance and interaction of all three conservative principles.

Mass – tells us why in certain regions flow accelerates.

Momentum – Relate the added circulation to the flow of the lift producing it.

Energy – allows us to relate the flow acceleration to the reaction force through Bernoulli's principle.

Common misconceptions:

- Longer path or Equal transit theory
  - Upper surface > lower surface
  - Velocity on the upper surface > velocity on the lower surface (to meet at the T.E.)
  - Pressure on the upper surface < Pressure on the lower surface
  - Net pressure ($\mathbf{P_{upper} - P_{lower}}$) → **Lift force**
  - Bernoulli's principle → $\mathbf{U_1 > U_2}$ → $\mathbf{P_1 < P_2}$ (comparison between two streamlines)
  - Bernoulli's principle is valid when we consider two points on two different streamlines when the flow is **irrotational** and **incompressible**.
- Skipping stone theory
  - The air pushes the foil from the bottom (**Not true**).
  - Air bounces off and creates a reaction force (**lift force**).
  - Airflow will not behave in this way.
  - There are a lot of forces generated between air particles that are not captured by this idea.

- **But from the theory** → The flow does change direction due to the presence of the foil and that redirection does take a force (lift).

- Venturi theory

At the leading edge of the airfoil, Rapid change in the curvature (**causes tight space**) causes acceleration (**venturi effect**).

According to the conservation of mass, the mass flow rate remains constant.

Flow sticks to the surface after the curvature at the leading edge due to the **Coanda effect**.

Flow leaves the trailing edge – **Kutta condition**